

Optimal Placement of PMUs using k -Power Domination Number in Grid Networks

J. Anitha^{1,*}, R. Manimegalai², S. Nithishkumar³, R. Arthi⁴

^{2,3,4} Department of Computer Science and Engineering

PSG Institute of Technology and Applied Research

Coimbatore-641 062

¹ Department of Mathematics

Rajalakshmi Institute of Technology

Chennai-600 124, India

Email:anithaharish78@gmail.com

Abstract

Electric power grids primarily rely on traditional meters, but they integrate more advanced and accurate Phasor Measurement Units (PMUs) to enhance monitoring and control capabilities. PMUs are deliberately selected for their ability to provide precise and synchronized readings of voltage and current at different points within the power grid, thereby playing a crucial role in grid operations. However, due to the high cost and the need for communication infrastructure, the installation of only a limited number of PMUs is possible in networks. Therefore, there is a need for a simple yet effective algorithm for Optimal PMU Placement (OPP) in electric grid systems. In this paper, k -power domination algorithm is used to identify nodes that are k -power dominating in the given network. The paper validates the k -power domination algorithm for specific networks and evaluates its performance across these networks. Furthermore, the algorithm's outputs are used for various other applications such as fault detection and localization, state identification, accuracy improvement, and cybersecurity enhancement, thereby making it a valuable tool for utilities and grid operators.

Keywords: Optimal PMUs Placement; Dominating Set; Facility Location Problem; Electrical Power System; Power Grid Monitoring.

1 Introduction

The placement of Phasor Measurement Units (PMUs) is viewed as an optimization problem where the goal is to find the lowest possible collection of PMUs to monitor the whole electric power system. A transmission line that connects two electrical nodes is represented by each edge in a graph representation $G(V, E)$ of the power system, where each vertex represents an electrical node. PMU placement with a view to measuring the state

*Corresponding Author

variable for the vertex at which it is put is the goal. Electricity firms need this monitoring in order to continuously evaluate the state of the system, which is determined by a number of crucial variables including the machine phase angle at generators and the magnitude of the voltage at loads.. The concept of k -power domination in graphs encapsulates the challenge of achieving comprehensive system monitoring with as few PMUs as possible. Despite recent cost reductions, comprehensive PMU deployment across all electric grid nodes is hampered by several challenges. As per industry protocols, PMUs are routinely placed in substations during refurbishment. However, strategic placement of PMUs at key buses is recommended to ensure system observability. This approach, when complemented with electric circuit analysis equipment, yields an economically feasible and technically sound solution.

Power dominance algorithms are utilized in this work to arrange PMUs in electrical grid networks in the best possible way. Any vertex in $V - S$ that has at least one neighbor in S is said to be dominating in a set $S \subseteq V$ of a graph $G(V, E)$. The domination number of G , $\gamma(G)$, represents the smallest size of such dominating sets. This idea is further developed by the power domination problem, which makes sure that every vertex in G is watched by set S , creating a Power Dominating Set (PDS). The symbol for the k -power dominance set is $\gamma_{p,k}(G)$. More extensive conditions are applied to the monitoring of the complete graph G . In order to achieve full observability of the electric power system k -power domination algorithm is employed. This approach not only enhances monitoring efficiency but also offers a cost-effective solution by minimizing the total number of PMUs required, thus overcoming the logistical, financial, and architectural challenges inherent in large-scale PMU deployment.

1.1 Preliminaries and Definitions

Path Graph: A path graph, also known as a linear graph, is a type of graph where nodes are arranged sequentially, forming a chain: $v_1, v_2, \dots, v_n$, with edges of the form $\{v_i, v_{i+1}\}$ where $i = 1, 2, \dots, n - 1$ [14].

Complete Graph: A complete graph, represented as K_n , is a graph in which each pair of unique vertices is connected by a single edge [14].

Cycle Graph: A cycle graph C_n , for $n \geq 3$, contains n edges and n vertices, with the edges forming a loop connecting the vertices in sequence from v_1 to v_n and back to v_1 [14].

Star Graph: In a star graph $K_{1,n}$, which is a tree with n nodes, one node has a vertex degree of $n - 1$, while the other $n - 1$ nodes each have a vertex degree of 1 [14].

Friendship Graph: A friendship graph, also called a fan graph F_n , is formed by joining N copies of the cycle graph C_3 with a common vertex, which acts as the universal vertex of the graph. The friendship graph F_n has $3n$ edges and $2n + 1$ vertices [14].

Ladder Graph: The ladder graph $L_{n,1} = P_n \times P_2$ is the Cartesian product of two path graphs, where one of the graphs has a single edge [14].

Binary Tree: A binary tree is a tree data structure where each node can have up to two children, typically called the left and right children. [16].

Cartesian Product of Graphs: The Cartesian product of graphs G and H , denoted as $G \times H$, possesses the following properties:

- In the Cartesian product graph $G \times H$, the vertices (u, v) and (u', v') are adjacent only if $u = u'$ and v is adjacent to v' in H , or $v = v'$ and u is adjacent to u' in G .

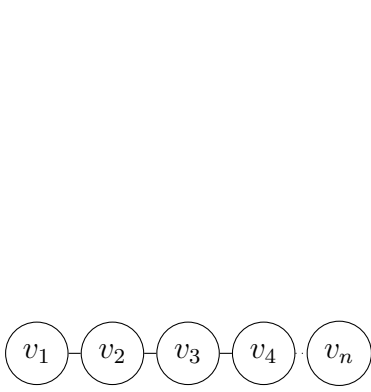
- The vertex set of $G \times H$ is given by the Cartesian product $V(G) \times V(H)$.

Dominating Set: In a graph G , a vertex v is referred to as a dominating set over a vertex u if u and v are neighboring vertices in G . A set $S \subseteq V(G)$ is called a dominating set if every vertex not in S is adjacent to at least one vertex in S . The minimum size of such a set S is called the domination number of G , denoted by $\gamma(G)$.

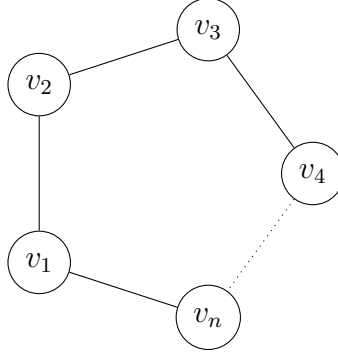
Power Dominating Set: Given an integer $k \geq 0$ and a graph $G(V, E)$, a set $S \subseteq V(G)$ is called a k -power dominating set if it monitors all vertices in G through a recursive process. Define the sets $M^i(S)$ of vertices monitored by S at level i as follows:

1. $M^0(S) = N[S]$
2. $M^{i+1}(S) = M^i(S) \cup \{N(v) : \exists v \in M^i(S), |N(v) \setminus M^i(S)| \leq k\}$.

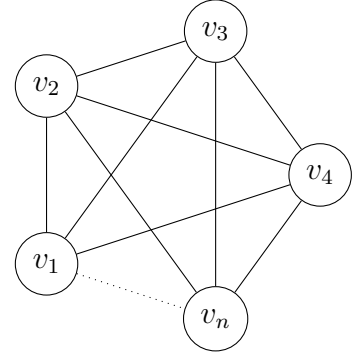
If $M^\infty(S) = V(G)$, then the set S is a k -power dominating set. The power domination number of G , denoted $\gamma_{p,k}(G)$, is the minimum size of a k -power dominating set in G .



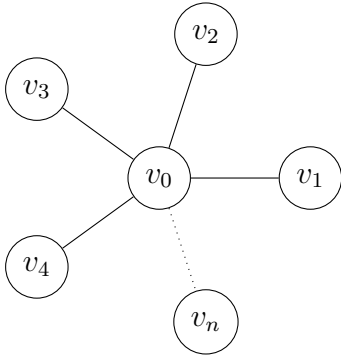
(a) Path Graph, P_n



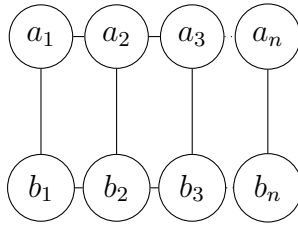
(b) Cycle Graph, C_n



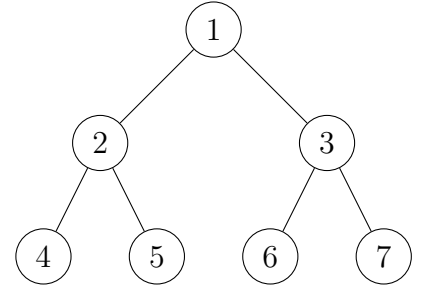
(c) Complete Graph, K_n



(d) Star Graph, $K_{1,n}$



(e) Ladder Graph, $L_{n,1}$



(f) Binary Tree, T_n

2 Literature review

It has been demonstrated that the power domination problem is NP-complete [1]. In fact, even when restricting the problem to chordal and bipartite graphs, it has been shown to

be NP-complete [1]. Lower and upper bounds on the power domination number for each graph G , $1 \leq \phi_p(G) \leq \gamma(G)$, are found in [1]. The issue of power dominance in trees is looked into [2]. Liao et al. (2005) demonstrated that the power domination number of split graphs, is an NP-complete problem. Additionally, if the interval ordering of the graph is known, a linear time algorithm and an $O(n \log n)$ time technique have been presented for determining the power domination number of an interval graph G .

For grids in [3], the power domination number is calculated. Zhao et al. looked at claw-free cubic graphs in 2006. Furthermore, it has been demonstrated that for each graph G of order $n \geq 3$, the power domination number satisfies $\gamma_p(G) < \frac{n}{3}$, and the extremal graphs that reached these upper bounds have been identified [4]. The goal of Xu et al.'s work [5] has been to develop a linear solution for handling the block graph power domination problem. The algorithm defines the extremal graphs and sets an upper boundary on the power domination number in block graphs. The color-marking based approach seeks to efficiently find a minimum power dominating set in block graphs so that a definite upper bound on the power domination number in this class of graphs may be established. Moreover, it described extremal graphs in this context as well. Still unexplored is the algorithm's flexibility in solving the power domination problem for various kinds of graphs that share tree-like structures, opening up new avenues for investigation and possible algorithmic improvements. Power dominance issues for generalized Petersen graphs [9], cylinders, torus, and product graphs [7] have all been thoroughly examined.

Chang et al. [8] present the concept of k -power domination, which is a general concept of dominance and power domination. The k -power domination number is an NP-complete problem as demonstrated by the computation of the problem's complexity in their work. [8] demonstrates the sharpness of the bound and shows that for any connected graph G of rank n , $\gamma_k^P(G) < \frac{n}{k+2}$. Additionally, it is demonstrated that for any $(k+2)$ -regular graph of rank n , $\gamma_k^P(G) < \frac{n}{k+3}$, and the graphs that meet this last restriction are described. The concept of propagation radius is presented by Dorbec et al. [12], defining it as the minimum number of steps required for the propagation of k power-dominating sets. The propagation radius for a significant range of parameters in Sierpinski graphs S_n is specifically determined in the paper. However, challenges are faced in establishing concise formulas, particularly for Sierpinski graphs S_n^p , where $2k+2 \geq p \geq k+1+\sqrt{k}+1$ and $n \geq 3$.

The computational complexity of failed power domination has been addressed by Glasser et al. [10], highlighting the NP-hardness of power domination computation. The paper focuses on identifying graphs where every vertex acts as its own power dominating set (PDS), contributing a list of such graphs. However, while offering this valuable insight, it fails to explore specific traits and properties of graphs with a power domination number of 0, which remains an uncharted domain.

Significant progress has been made by Brimkov et al. [11] in exploring connected power domination within graphs, with the NP-hardness of determining the domination number in connected graphs being firmly established. NP-completeness of certain graphs has been proven, leading to various structural discoveries. Various structural insights have been provided by their research through the proof of the NP-completeness of specific graphs. Moreover, a formula for the biconnected components' numerical values and the graph's connected power dominance number have been determined. Despite these successes, attempts to determine the connected power domination of IEEE Bus 300 have run into problems since the calculations have exceeded timeout limits, revealing problems with the scalability of the computations and their practical use.

The employment of bounded-treewidth dynamic programs to solve the power dominating set has been the main focus of Guo et al. [15]. [15] presents a linear-time technique that is streamlined for determining the power dominant set in trees. Furthermore, it has been shown that the power dominating set, as defined by $|P|$, is not more accurately measurable than the dominating set and is $W[2]$ -hard. However, there are still a lot of topics that require additional study. It is yet unknown if the power dominating set contains nontrivial polynomial-time data reduction methods similar to those we have for the dominating set, given the complexity of the two sets differs greatly.

3 k -Power Domination Algorithm

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Algorithm PDS( $G(V, E), k$ )
{
   $min\_set \leftarrow V(G)$ 
   $min\_set\_size \leftarrow \infty$ 
   $vertices \leftarrow$  list of all vertices in  $G$ 
  for  $size := 1$  to  $|vertices|$  {
    for each subset  $S \subseteq vertices$  with  $|S| = size$  {
       $monitored \leftarrow S \cup (\bigcup_{v \in S} N(v))$ 
       $new\_monitored \leftarrow monitored$ 
      while True {
         $to\_add \leftarrow \emptyset$ 
        for each vertex  $v \in monitored$  {
           $neighbors \leftarrow N(v)$ 
           $unmonitored\_neighbors \leftarrow neighbors \setminus monitored$ 
          if  $|unmonitored\_neighbors| \leq k$  {
             $to\_add \leftarrow to\_add \cup unmonitored\_neighbors$ 
          }
        }
      }
      if  $to\_add = \emptyset$  {
        break
      }
       $new\_monitored \leftarrow new\_monitored \cup to\_add$ 
       $monitored \leftarrow new\_monitored$ 
    }
    if  $|monitored| = |V(G)|$  {
      if  $|S| < min\_set\_size$  {
         $min\_set \leftarrow S$ 
         $min\_set\_size \leftarrow |S|$ 
      }
    }
  }
}
return  $min\_set$ 
}

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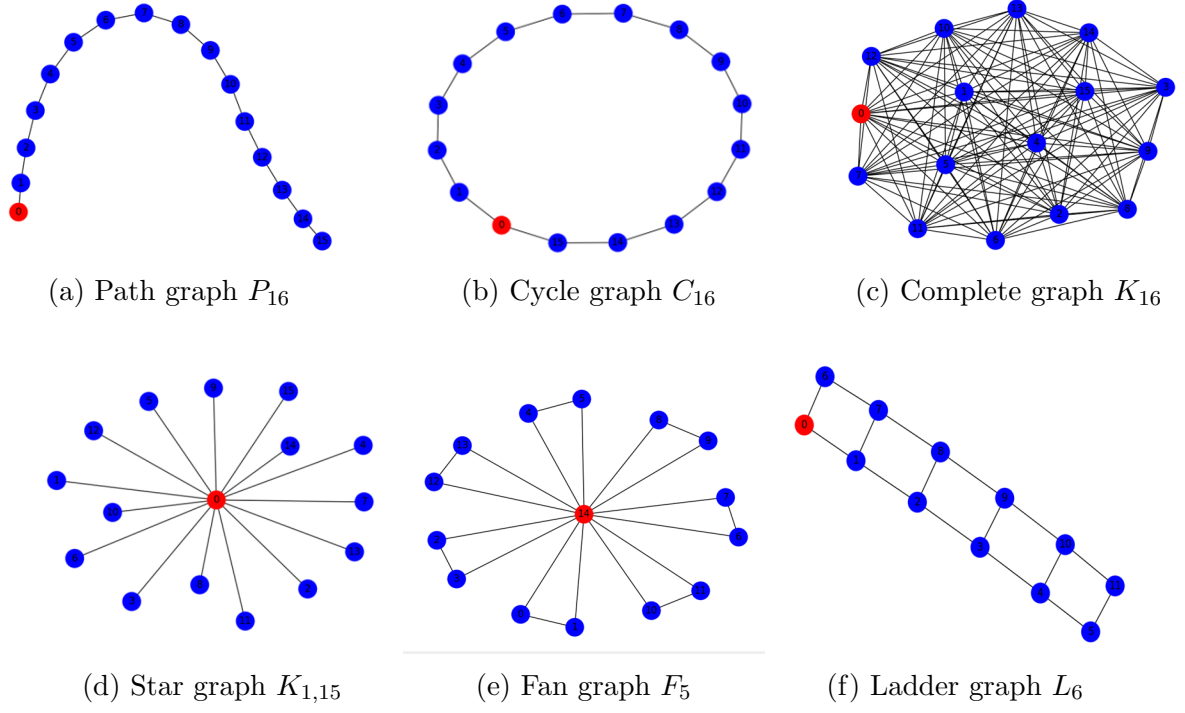


Figure 2: k -Power Dominating Set in Different Types of Graph Networks using k -Power Domination Algorithm

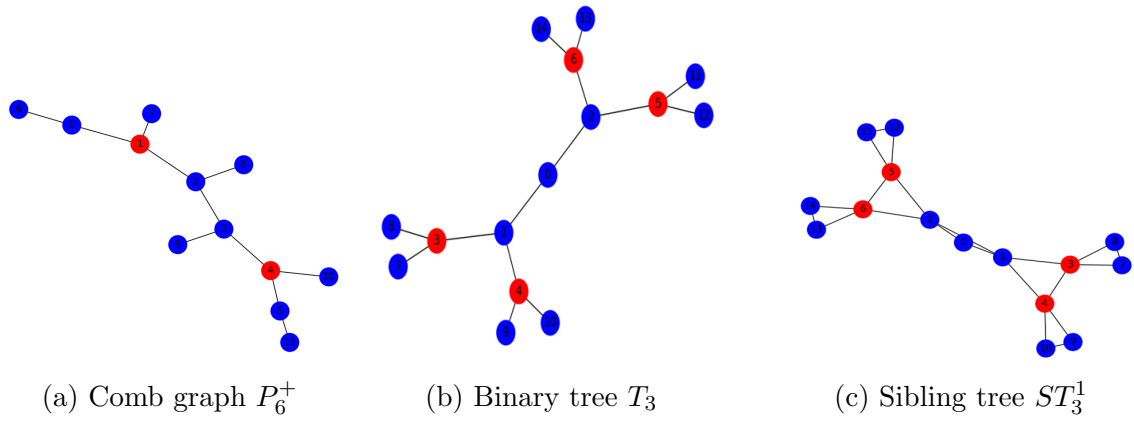


Figure 3: Power Dominating Set in Comb Graph, Binary Tree and Sibling Tree using Power Domination Algorithm

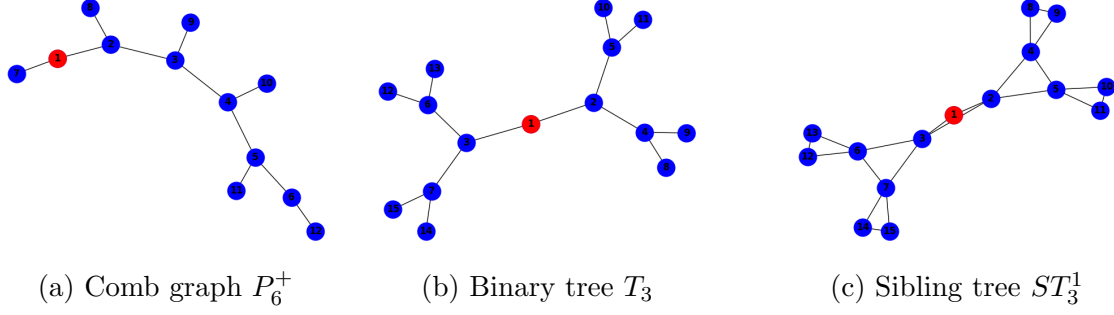


Figure 4: 2-Power Dominating Set in Comb Graph, Binary Tree and Sibling Tree using 2-Power Domination Algorithm

The PDS algorithm identifies the k -power domination set, depicted by red vertices in the pictures above, as the smallest subset of nodes required for full network observability. The k -power domination set is deliberately placed in each graph to provide optimal coverage, whether along linear lines, across cyclic structures, at central hubs and outlying nodes, or scattered across multiple layers, improving power system reliability and stability. For binary, sibling, and comb graphs, the power domination algorithm differs significantly from the 2-power domination algorithm, which adds redundancy. However, for other graphs, both $k=1$ and $k=2$ yield the same result, demonstrating that the smallest subset is sufficient for comprehensive coverage.

Table 1 shows the execution time to compute the k -power dominating set for various types of graphs with 16 nodes

Type of Graph	Execution time	
	k=1	k=2
Path P_n	1.018	1.020
Cycle C_n	0.980	0.945
Ladder L_n	0.866	0.929
Complete graph K_n	0.971	0.950
Star Graph $K_{1,n}$	1.010	1.085
Fan graph F_n	0.131	0.129
Comb graph P_n^+	0.415	0.447
Binary tree T_n	1.85	2.12
Sibling tree ST_n^1	0.477	0.379

Table 1: Execution Time to Compute the k -Power Dominating Set for Graphs with 16 Nodes

3.1 Optimal PMU Placement

The proposed k -power domination algorithm is a highly efficient and robust solution for the optimal placement of Phasor Measurement Units (PMUs) in electric grid systems. It ensures comprehensive network observability by identifying a minimal subset of nodes where PMUs should be placed, thereby guaranteeing that each node in the grid is monitored either directly or indirectly. This approach not only minimizes the number of PMUs

required but also maximizes coverage, enhancing the reliability and stability of the electric grid. Unlike traditional greedy algorithms, which often necessitate more PMUs to achieve full observability and may not always yield minimal sets, the k -power domination algorithm provides a precise and minimal solution. Compared to Integer Linear Programming (ILP) methods, which can be computationally intensive and impractical for large-scale networks, this algorithm is computationally efficient and scalable, making it suitable for various grid topologies and sizes.

Moreover, the k -power domination algorithm incorporates the concept of k -redundancy, ensuring that each node is observed by multiple PMUs, which significantly improves fault tolerance and robustness against failures. This redundancy guarantees continuous monitoring and control even in the event of PMU malfunctions. The algorithm's flexibility allows it to adapt to dynamic changes in network topology, such as the addition or removal of nodes, ensuring sustained optimal performance in evolving grid environments.

In terms of cost efficiency, the k -power domination algorithm reduces the overall number of PMUs needed, thereby lowering implementation and maintenance costs. By optimizing PMU placement to minimize overlap and maximize observability, it ensures operational efficiency and effective data collection. Its proven effectiveness in various studies and real-world applications highlights its practical applicability, making it a preferred choice for grid operators aiming to enhance monitoring and control of electric grids. Overall, the k -power domination algorithm represents a superior method for PMU placement, addressing the limitations of other approaches and significantly contributing to the reliability, stability, and efficiency of power systems.

The following Table II depicts the cost of placement of PMUs in electric grid networks by implementing k -power domination algorithm.

Type of Network	Heuristic methods	PDS Algorithm	
		$k = 1$	$k = 2$
Path P_n	300000	50000	50000
Cycle C_n	300000	50000	50000
Ladder L_n	250000	50000	50000
Complete graph K_n	50000	50000	50000
Star Graph $K_{1,n}$	50000	50000	50000
Fan graph F_n	50000	50000	50000
Comb graph P_n^+	300000	100000	50000
Binary tree T_n	350000	200000	50000
Sibling tree ST_n^1	250000	200000	50000

Table 2: Cost of placement of PMUs using Various Strategies(in dollars)

The Table II shows costs associated with placing Phasor Measurement Units (PMUs) using heuristic methods and the Power Dominating Set (PDS) algorithm across various network topologies. It demonstrates that the PDS algorithm is generally more cost-effective than heuristic methods. For Path p_n , Cycle C_n , Ladder L_n , and Comb graph P_n^+ , the heuristic method costs higher compared to the PDS algorithm's \$50,000. This trend is similar for Binary Tree (Tn) and Sibling Tree (ST) topologies, where the heuristic methods cost \$350,000 and \$250,000 respectively, while the PDS algorithm costs \$200,000. For $k = 2$, both methods generally show similar costs, around \$50,000, except for Binary Tree (Tn) and Sibling Tree (ST), where the heuristic methods still have higher costs compared to the PDS algorithm's \$50,000. This analysis indicates that the PDS algorithm

effectively reduces the cost of PMU placement, especially for more complex or larger networks, by optimizing the number and placement of PMUs needed for comprehensive system observability.

4 Conclusions

In this paper, Power domination algorithm is used to improve grid monitoring and control systems. The scalability of the power domination and 2-power domination algorithm is tested for various graph structures. Despite the challenges encountered with specific configurations such as ladder graphs, the algorithm demonstrates its effectiveness in graphical networks and adaptability to structural context. This adaptability is important for real-world applications, where power grids often have complex and heterogeneous structures. In order to control and ensure the stability and reliability of modern power systems, PMUs play central role by providing accurate and timely measurements of voltage and current on the network. Network operators gain comprehensive, real-time information about system status, allowing them to detect and respond to incidents, interruptions and errors more effectively by strategically placing PMUs in key locations.

As the scale and complexity of power grid increases, efficient algorithms are necessary to ensure the feasibility and efficiency of PMU deployment. High computational cost and increasing network size demands the need for solutions that can scale to large-scale network systems without compromising the performance. Prioritizing scalability in algorithm design enables the development of PMU placement strategies. Leveraging the strengths of algorithm such as the k -power domination algorithm and prioritizing scalability in algorithm design helps in developing effective and efficient PMU placement strategies tailored to needs of modern electric grid. Continued research and innovation in this area is essential to build reliable, resilient and sustainable energy infrastructure that meets the changing needs of electric power.

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